UNIT–2 AGILE MINI-PROJECT

Student Record and Academic Management Platform

Submitted By

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Agile / Scrum Project Planning Report

This mini-project is prepared according to the Unit–2 Agile project activities and deliverables.

1. Business Problem and Product Vision

The Student Record and Academic Management Platform is designed to help educational institutions manage student-related academic information in an organized and efficient way. The system covers student records, attendance, academic performance, course registration, authentication, notifications, and reporting.

The Agile planning process begins by understanding stakeholders, business requirements, functional and non-functional requirements, user personas, the Minimum Viable Product (MVP), and the product roadmap.

Product Vision

Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration, and reporting through a scalable web-based application.

2. Agile Product Vision

• Overall objective: Provide a centralized platform for managing student academic information.

• Primary users: Students, faculty members, administrators, and other authorized academic staff.

• Business benefits: Reduced manual work, better record accuracy, faster reporting, and improved academic monitoring.

• Success criteria: Core modules should work correctly, records should be easy to update and retrieve, and authorized users should access appropriate features.

• Expected capabilities: Student management, course management, attendance, marks, authentication, notifications, and reports.

3. Scrum Roles

Role	Responsibilities
Product Owner	Defines requirements, prioritizes the Product Backlog, represents stakeholder interests, and accepts completed work.
Scrum Master	Facilitates Scrum ceremonies, removes impediments, ensures Agile practices are followed, and supports collaboration.
Development Team	Designs the application, writes code, tests software, prepares documentation, and demonstrates completed features.

4. Project Epics

Epic	Main Features
1. Student Management	Student registration, profile management, search, update, and deletion.
2. Course Management	Add course, update course, assign students, and view course details.
3. Attendance Management	Record attendance, view attendance, and generate attendance reports.
4. Marks & Academic Performance	Enter/update marks, calculate grades, and generate result sheets.
5. Authentication & Security	Login, logout, password reset, and role-based access control.
6. Notifications	Attendance alerts, examination notifications, and result notifications.
7. Reporting	Student, attendance, performance, and course reports.

5. User Stories and Acceptance Criteria

User Story 1 – Student Login: As a registered user, I want to log in securely so that I can access the appropriate dashboard and academic services.

• Login page loads successfully.

• Registered users can log in.

• Invalid credentials display an appropriate error message.

• Successful login redirects to the dashboard.

• Session management functions correctly.

User Story 2 – Record Attendance: As a Faculty Member, I want to record student attendance so that attendance records remain accurate and updated.

• Faculty can select a course.

• Faculty can select a lecture date.

• Faculty can mark attendance.

• Attendance records are saved successfully.

• Attendance reports can be generated.

User Story 3 – Add Student: As an Administrator, I want to add a student so that the student's academic record can be maintained.

• Student details can be entered.

• Required fields are validated.

• Student record is saved successfully.

• The new student can be searched after creation.

6. Prioritized Product Backlog

Priority	Epic	User Story	Story Points
High	Authentication	User Login	3
High	Student Management	Add Student	3
High	Student Management	Search Student	2
High	Attendance	Record Attendance	5
Medium	Course Management	Add Course	3
Medium	Reporting	Attendance Report	5
Medium	Notifications	Attendance Alert	5
Low	Dashboard	Academic Analytics	8

7. Story Point Estimation

Story points are used to estimate relative effort and complexity. The scale used in this project is: 1 = Very Easy, 2 = Easy, 3 = Moderate, 5 = Difficult, 8 = Very Difficult, and 13 = Highly Complex.

8. Sprint Planning

Sprint	Goal	Activities

Sprint 1	User authentication and project setup	Repository initialization, user registration, login, basic UI, and testing.
Sprint 2	Student Management Module	Add student, search student, update student, and delete student.
Sprint 3	Course and Attendance Management	Add course, assign courses, record attendance, and attendance reports.

9. Scrum Board

The Scrum Board is used to track work throughout the sprint.

Backlog	To Do	In Progress	Testing	Completed
Login	Add Student	Search Student	Reports	Project Setup
Course	Attendance	Notifications		

10. Scrum Ceremonies

Sprint Planning: Select user stories, estimate effort, and define the Sprint Goal.

Daily Scrum: Each member discusses what was completed, what will be done next, and any challenges.

Sprint Review: Demonstrate completed features and collect stakeholder feedback.

Sprint Retrospective: Discuss what went well, challenges faced, and improvements for the next sprint.

11. Agile Framework Comparison

Framework	Main Idea	Suitability
Scrum	Iterative development using roles, events, and a Product Backlog.	Highly suitable for this student project.
Kanban	Visualizes continuous flow of work using a board.	Useful for tracking ongoing tasks.
Extreme Programming (XP)	Focuses strongly on software quality, testing, and frequent feedback.	Useful for development practices.
Lean Development	Reduces waste and focuses on delivering value.	Useful for process improvement.
LeSS	Scales Scrum for multiple teams.	More suitable for larger projects.
SAFe	Provides a structured approach for scaling Agile across large organizations.	More suitable for enterprise-scale projects.
Scrum@Scale	Coordinates multiple Scrum teams while retaining Scrum principles.	Useful when many Scrum teams are involved.

12. Sprint Review Report

At the Sprint Review, the team demonstrates completed features such as project setup, authentication, student management, course management, and attendance functionality. Feedback is recorded and considered for the next sprint.

13. Sprint Retrospective Report

• What went well: Tasks were divided into manageable user stories and prioritized.

• Challenges: Some features may require additional testing and coordination.

• Improvement: Improve task estimation, communication, testing, and documentation in the next sprint.

14. Project Timeline and Release Roadmap

Phase 1 – Project setup and authentication → Phase 2 – Student Management → Phase 3 – Course and Attendance → Phase 4 – Marks, Notifications and Reporting → Phase 5 – Testing, documentation and release.

15. Conclusion

This mini-project applies Agile and Scrum concepts to the Student Record and Academic Management Platform. The project is divided into clear epics, user stories, acceptance criteria, backlog items, story points, and sprints. Using Scrum helps the team deliver the system incrementally, receive feedback regularly, and improve the product throughout development.

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